

1. According to the 2018 National Waste Policy, Australians produce 540kg of household waste (per family), each year. Assume the standard deviation of the distribution is 100kg. Find the probability that the mean of a sample of 55 families will be between smaller than 500kg. Write a decimal number with three decimal places.

Solution: For the distribution of the sample means, $\mu = 540$, while $\sigma = \frac{100}{\sqrt{55}} = 13.484$.

We want $P(X \leq 500)$.

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pnorm(500,540,13.484)
#0.001506156
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2. Urn I and urn II each have two red chips and two white chips. Two chips are drawn simultaneously from each urn. Let X_1 be the number of red chips in the first sample and X_2 the number of red chips in the second sample. Find the probability that $P(X_1 + X_2 \geq 3)$. Write a decimal number with three decimal places.

Solutions:

$$p_{X_1}(k) = p_{X_2}(k) = \frac{\binom{2}{k} \binom{2}{2-k}}{\binom{4}{2}} \quad k = 0, 1, 2$$

For $X_1 + X_2 = m$, let $X_1 = k$ and $X_2 = m - k$ for $k = 0, 1, \dots, m$. Then $p_{X_1+X_2}(m) = \sum_{k=1}^m p_{X_1}(k)p_{X_2}(m-k)$ with $m = 0, 1, 2, 3, 4$. Then

m	$p_{X_1+X_2}(m)$
0	1/36
1	2/9
2	1/2
3	2/9
4	1/36

Then $P(X_1 + X_2 \geq 3) = \frac{2}{9} + \frac{1}{36} = 0.25$.

3. Let X and Y be two continuous random variables defined over the unit square. What does c equal if their joint density is $f_{XY}(x, y) = c(x^2 + y^2)$?

Solution:

$$\begin{aligned}
 1 &= \int_0^1 \int_0^1 c(x^2 + y^2) dx dy \\
 &= c \int_0^1 \int_0^1 x^2 dx dy + c \int_0^1 \int_0^1 y^2 dx dy \\
 &= c \int_0^1 \frac{x^3}{3} \Big|_0^1 dy + c \int_0^1 \frac{y^3}{3} \Big|_0^1 dx \\
 &= c \int_0^1 \frac{1}{3} dy + c \int_0^1 \frac{1}{3} dx \\
 &= \frac{2}{3}c
 \end{aligned}$$

then $c = \frac{3}{2} = 1.5$.

4. Let $f_Y(y) = \frac{3}{14}(1 + y^2)$ for $0 \leq y \leq 2$. Define the random variable W by $W = 3Y + 2$. Compute $f_W(4)$. Write a decimal number with three decimal places.

Solution:

$$f_W(w) = \frac{1}{|3|} \cdot \frac{3}{14} \left[1 + \left(\frac{w-2}{3} \right)^2 \right] = \frac{1}{126} (w^2 - 4w + 13)$$

with $0 \leq y \leq 2$ implies $0 \leq 3y \leq 6$ or $2 \leq 3y + 2 \leq 8$ and then $2 \leq w \leq 8$.

Therefore $f_W(4) = 0.1031746$.

5. In the setting of the previous question, compute $f_W(1)$. Write a decimal number with three decimal places.

Solution: Since $2 \leq w \leq 8$, $f_W(1) = 0$.

6. Marksmanship competition at a certain level requires contestant to take ten shots with each of two different handguns. Final scores are computed by taking the weighted average of four times the number of bull's-eyes made with the first gun plus six times the number gotten with the second. If Cathie has a 30% chance of hitting the bull's-eye with each shot from the first gun and a 40% chance with each shot from the second gun, what is her expected score? Consider shots independent. Write an integer.

Solution:

Let $X_i = 1$ is a shot with the first gun is a bull's eye and 0 otherwise. $E(X_i) = 0.30$.

Let $V_i = 1$ if a shot with the second gun is a bull's eye and 0 otherwise.
 $E(V_i) = 0.40$.

Cathie's score is

$$4 \sum_{i=1}^{10} X_i + 6 \sum_{i=1}^{10} V_i$$

and her expected score is

$$E \left(4 \sum_{i=1}^{10} X_i + 6 \sum_{i=1}^{10} V_i \right) = 4 \cdot 10 \cdot 0.30 + 6 \cdot 10 \cdot 0.40 = 36$$

7. Let $Y_1, Y_2, \dots, Y_{10}$ be a random sample from the exponential pdf $f_Y(y) = 2e^{-2y}$, $y \geq 0$. What is $P(Y_{\min} < 0.2)$? Write a decimal number with three decimal places.

Solution: We have that

$$P(Y_{\min} < 0.2) = \int_0^{0.2} n f_Y(y) [1 - F_Y(y)]^{n-1} dy$$

and $F_Y(y) = 1 - e^{-2y}$.

Then

$$\begin{aligned} \int_0^{0.2} 10 f_Y(y) [1 - F_Y(y)]^9 dy &= \int_0^{0.2} 10 \cdot 2e^{-2y} [1 - 1 + e^{-2y}]^9 dy \\ &= 20 \int_0^{0.2} e^{-20y} dy \\ &= 20 \left(-\frac{1}{20} \right) [e^{-20y}]_0^{0.2} \\ &= -e^{-20 \cdot 0.2} + 1 \\ &= 1 - e^{-20 \cdot 0.2} = 0.9816844 \end{aligned}$$

8. Suppose a die is rolled six times. Let X be the total number of 4's that occur and let Y be the number of 4's in the first two tosses. Compute $p_{Y|X=3}(2)$. Write a decimal number with three decimal places.

Solutions: The probability that $X = x$ and $Y = y$ is the probability of y 4's on the first rolls and $x - y$ rolls on the last four rolls. These events are independent, so

$$p_{XY}(x, y) = \binom{2}{y} \left(\frac{1}{6} \right)^y \left(\frac{5}{6} \right)^{2-y} \binom{4}{x-y} \left(\frac{1}{6} \right)^{x-y} \left(\frac{5}{6} \right)^{4-x+y}$$

for $y \leq x$.

Then

$$p_{Y|x} = \frac{p_{XY}(x, y)}{p_X(x)} = \frac{\binom{2}{y} \left(\frac{1}{6}\right)^y \left(\frac{5}{6}\right)^{2-y} \binom{4}{x-y} \left(\frac{1}{6}\right)^{x-y} \left(\frac{5}{6}\right)^{4-x+y}}{\binom{6}{x} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{6-x}} = \frac{\binom{2}{y} \binom{4}{x-y}}{\binom{6}{x}}$$

for $0 < y < \min(2, x)$, which is a hypergeometric distribution.

Therefore

$$p_{Y|X=3}(2) = \frac{\binom{2}{2} \binom{4}{1}}{\binom{6}{3}} = 0.2$$

9. Electromagnetic fields generated by power transmission lines are suspected by some researchers to be a cause of cancer. Especially at risk would be workers of telephone companies because of their frequent proximity to high-voltage wires. According to one study, 2 cases of a rare form of cancer were detected among a group of 9500 telephone workers. In the general population, the incidence of that particular condition is one case in a 100,000 people. What is the probability to observe at least 2 cases using the Poisson approximation to the binomial distribution? Write a decimal number with three decimal places.

Solution:

We have that $n = 9,500$ and $p = 1/100,000$, then $np = 0.095$ and

$$P(X \geq 2) = 1 - P(X \leq 1)$$

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1-ppois(1,n*p)
[1] 0.004236637
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10. Consider $Z \sim N(0,1)$. Compute $P(0 \leq Z \leq 1.24)$? Write a decimal number with three decimal places.

Solution:

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pnorm(1.24)-pnorm(0)
[1] 0.3925123
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